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MILITARY SPECIFICATION

CABLE, MISSILE CONTROL, TORPEDO TUBE, MK 8 MOD 0 AND MK 9 MOD 0

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements for the procurement of missile control cable assemblies for SUBROC missiles in accordance with NAVSEA LD 162872 and/or LD 162873.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-B-117	Bags, Sleeves and Tubing, Interior Packaging
MIL-S-901	Shock Tests, H. I. (High Impact); Shipboard Machinery, Equipment and Systems; Requirements for
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for
MIL-C-82621	Cable, Electric, Missile, 54-Conductor

FSC 1355

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STANDARDS

Federal

Federal Test Method Standard No. 228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-type Base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Naval Sea Systems Command (Code Ident. 10001)

LD 162872	Cable, Missile Control, Mk 8 Mod 0
LD 162873	Cable, Missile Control, Mk 9 Mod 0
LD 162874	Connector, Torpedo Tube Cable, Mk 47 Mod 0
LD 162875	Connector, Torpedo Tube Cable, Mk 48 Mod 0
Dwg 757989	Container
Dwg 2865910	In-Line Connector

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The missile control cable assembly consists of a tube door plug, a 54-conductor electric cable, and an in-line connector.

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3.1.1 Tube door plug. The tube door plug mates with Torpedo Tube Cable Connector Mk 47 Mod 0 (NAVSEA LD 162874) or Mk 48 Mod 0 (NAVSEA LD 162875). The plug consists of a machined aluminum casting, a polycarbonate insulator, and copper alloy contacts. The contacts are crimped to a 54-conductor cable. A trigger-type latch holds the plug properly mated.

3.1.2 Cable. The 54-conductor electric cable shall be as described in MIL-C-82621.

3.1.3 In-line connector. The in-line connector, NAVSEA Dwg 2865910, mates with the connector on the console cable.

3.2 General

3.2.1 NAVSEA drawings. In addition to the requirements specified herein, the missile control cable shall conform to NAVSEA LD 162872 and/or LD 162873 and all drawings and specifications listed therein.

3.2.2 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Classification of requirements. The requirements for the cable assemblies are classified herein as follows:

Requirement	Paragraph
Preproduction	3.4
Materials	3.5
Design and construction	3.6
Mechanical requirements	3.7
Electrical requirements	3.8
Environmental requirements	3.9

3.4 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the cable is required. The sample shall be manufactured using the methods and procedures proposed for the production lot. This sample will be tested as specified in 4.3.1 and is for the purpose of determining, prior to starting production, that the manufacturer's production process will produce cables that will meet the requirements of this specification.

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3.5 Materials. All materials shall be as specified herein and in accordance with NAVSEA LD 162872 and/or LD 162873.

3.6 Design and construction. The design and construction shall be as defined on NAVSEA LD 162872 and/or LD 162873.

3.7 Mechanical requirements. The cable assembly shall successfully meet the following mechanical requirements:

3.7.1 Mating or unmating force. The tube door plug shall fully engage or disengage with an axial force of not more than 60 pounds when tested in accordance with 4.5.2.

3.7.2 Cable strength. The cable shall withstand a 200-pound load without functional failure in accordance with 4.5.3.

3.7.3 Cable flexibility. The cable shall withstand a torsion test and a low temperature bend test without functional failure in accordance with 4.5.4.

3.8 Electrical requirements. The cable assembly shall successfully meet the following electrical requirements:

3.8.1 Dielectric strength. The cable assembly shall withstand, without arcing, a potential of 1000 volts ac when tested in accordance with 4.5.5.1.

3.8.2 Insulation resistance. The insulation resistance of the missile control cable assembly shall be greater than 70 megohms when measured with a potential of 500-750 vdc in accordance with 4.5.5.2.

3.8.3 Continuity. The missile control cable assembly shall indicate continuity at each of the 54 contacts when tested as in 4.5.5.3.

3.8.4 Contact, conductor resistance. The maximum resistance of each circuit shall be as shown in Table I when subjected to the test defined in 4.5.5.4.

TABLE I

Conductor AWG Size	Maximum DC Resistance
18	0.4 ohm
12	0.3 ohm

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3.9 Environmental requirements. The missile control cable assembly shall be subjected to the following environmental conditions and shall meet the requirements as specified herein:

3.9.1 Vibration. The cable assembly shall successfully withstand a vibration test in accordance with MIL-STD-167, Type I, and 4.5.6.1 of this specification. During this test there shall be no circuit discontinuities exceeding 100 microseconds in duration. At the conclusion of the test, the unit shall meet the requirements of 3.8.2 (Insulation resistance) and 3.8.3 (Continuity).

3.9.2 Shock. The missile control cable assembly, mounted to simulate service installation, shall be subjected to the shock tests of MIL-S-901 for light weight equipment in accordance with 4.5.6.2. Following this test, the assembly shall meet the requirements of 3.8.2 and 3.8.3.

3.9.3 Salt water immersion. The missile control cable assembly, with connector attached, shall successfully withstand a salt water immersion test in accordance with 4.5.6.3. Following this test, the assembly shall meet the requirements of 3.8.2 and 3.8.3.

3.9.4 Low temperature (operating). The cable assembly shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3 during the test defined in 4.5.6.4.

3.9.5 High temperature (operating). The cable assembly shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3 during the test defined in 4.5.6.5.

3.9.6 Temperature and humidity (operating). The cable assembly shall be subjected to the temperature and humidity (operating) test of 4.5.6.6, after which it shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3.

3.9.7 Temperature and humidity (storage mode). The missile control cable shall be subjected to the temperature and humidity test of 4.5.6.7 and thereafter shall meet the requirements of 3.8.2 and 3.8.3.

3.9.8 Tropical environment. The missile control cable shall be subjected to the tropical environment of 4.5.6.8 and shall meet the requirements of 3.8.2 and 3.8.3.

3.10 Workmanship. The cable assemblies furnished to this specification shall be constructed and finished in a manner to assure full compliance with all requirements of this specification and applicable NAVSEA drawings. Particular attention shall be paid to the neatness and quality of soldering, wiring, and freedom from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 Preproduction. Four Mk 8 Mod 0 or four Mk 9 Mod 0 cable assemblies shall be supplied by the contractor (if both types are being supplied, two of each is sufficient). The cable assemblies shall be manufactured using the same procedures and methods and at the same plant location proposed by the contractor for the execution of the contract. The sample shall be submitted to the procuring activity for purposes of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the item by the contractor prior to the approval of the procuring activity or completion of inspection and tests on the preproduction sample shall be at the contractor's risk. Accepted preproduction samples will become the property of the procuring activity and will be included in the number of cable assemblies called for in the contract or order.

4.2.2 Acceptance. Acceptance tests as specified herein shall be conducted on every cable assembly.

4.3 Inspection

4.3.1 Preproduction testing and acceptance testing. Preproduction testing and acceptance testing shall consist of the inspections and tests of 4.4.1 and 4.4.2 on the samples selected under 4.2.1 and 4.2.2, as applicable.

4.3.1.1 Action in case of failure.

4.3.1.1.1 Preproduction sample. In case of failure in any preproduction sample, the cause shall be determined and corrected. A statement of the failure and proposed corrective action shall be submitted to the procuring activity for approval before additional tests are conducted.

4.3.1.1.2 Acceptance tests. In case of failure in acceptance tests, the cause shall be determined and a statement of the failure and proposed corrective action shall be submitted to the procuring activity for

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consideration and approval. No subsequent testing shall be conducted without the approval of the procuring activity.

4.4 Tests

4.4.1 Preproduction tests. The preproduction test requirements are shown in Table II. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity.

4.4.2 Acceptance tests. The manufacturer shall maintain a 100 percent inspection procedure to assure that each cable assembly meets the requirements of Table III.

4.4.3 Test conditions. Unless otherwise specified, tests and examinations required by this specification shall be conducted under any combination of conditions within the ranges below. Any specified condition shall not affect the other two ambient ranges.

(a) Temperature: 20 degrees C to 30 degrees C.

(b) Relative humidity: Room ambient to 95 percent.

(c) Barometric pressure: 24 to 31 inches of mercury.

4.5 Test methods

4.5.1 Examination of product. The cable assemblies and components shall be examined to insure conformance to all requirements of this specification and the applicable NAVSEA drawings. Assurance shall be made that no changes have been made in the cable assemblies other than those granted for preproduction approval. Examination shall be performed to assure compliance with the following requirements:

(a) 3.2.1 NAVSEA drawings.

(b) 3.5 Materials.

(c) 3.6 Design and construction.

(d) 3.10 Workmanship

4.5.2 Connector mating and unmating force. A torpedo tube cable connector, NAVSEA LD 162874 or LD 162875, shall be mounted in a movable holding device capable of measuring force in both a push and pull direction. The appropriate missile cable assembly tube door plug shall be engaged and disengaged to the mating connector by hand as in service use. The force required to engage and disengage shall be not greater than 60 pounds. See Figure 1.

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TABLE II. PREPRODUCTION TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
Acceptance Tests	4.4.2	4.4.2
Cable Strength	3.7.2	4.5.3
Cable Flexibility	3.7.3	4.5.4
Dielectric Strength	3.8.1	4.5.5.1
Vibration	3.9.1	4.5.6.1
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Shock	3.9.2	4.5.6.2
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Salt Water Immersion	3.9.3	4.5.6.3
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Low Temperature (Operating)	3.9.4	4.5.6.4
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Mating and Unmating Force	3.7.1	4.5.2
High Temperature (Operating)	3.9.5	4.5.6.5
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Mating and Unmating Force	3.7.1	4.5.2
Temperature & Humidity (Operating)	3.9.6	4.5.6.6
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Mating and Unmating Force	3.7.1	4.5.2
Temperature/Humidity (Storage Mode)	3.9.7	4.5.6.7
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3
Tropical Environment	3.9.8	4.5.6.8
Insulation Resistance	3.8.2	4.5.5.2
Continuity	3.8.3	4.5.5.3

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TABLE III. INDIVIDUAL TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2.1, 3.5, 3.6, 3.10	4.5.1
Dielectric Strength	3.8.1	4.5.5.1
Insulation Resistance	3.8.2	4.5.5.2
Connector Mating and Unmating Force	3.7.1	4.5.2
Contact, Conductor Resistance	3.8.4	4.5.5.4

4.5.3 Cable strength. Securely mount a torpedo tube cable connector, NAVSEA LD 162874 or LD 162875, in a suitable fixture (Figure 2). Mate the appropriate missile control cable tube door plug with the connector and securely clamp the cable in a horizontal plane 15 plus 1 minus 0 inches from the connector. Attach the weight support to the cable at a point midway between the plug and cable clamp. The weight support shall be no more than 1.5 inches long and of such design as to spread the load evenly over the 1.5 inch length of cable. The cable sheath may be protected or padded to prevent damage. A weight of 200 pounds shall be suspended from the cable for a period of at least 60 seconds, not including the application and removal periods. During application and removal of this force, all 54 circuits shall be monitored for a discontinuity of greater than 100 microseconds. The circuits may be wired in series. The load shall be applied and released a total of 20 times over a period not to exceed 2 hours.

4.5.4 Cable flexibility

4.5.4.1 Torsion test

(a) While it is mounted in a suitable test fixture, a torsion test shall be performed on the missile control cable assembly. The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 4 inches shall be formed within a 10-inch length of cable (see Figure 3). The cable shall be rigidly secured to a fixed support at one clamp position, and rigidly secured to a pivot arm at the second clamp position. The pivot axis of the arm shall be an extension of the centerline of the cable which is clamped to the fixed support.

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(b) One cycle of torsion shall consist of a 90-degree rotation in one direction and a 180-degree rotation in the reverse direction, and a 90-degree rotation in the original direction to the start position, in a plane perpendicular to the fixed support.

(c) A test shall be conducted at a rate of 2 cycles per minute. The time duration of one cycle shall approximate but not exceed 15 seconds. Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 3000 cycles of torsion.

4.5.4.2 Low temperature (90-degree bend test)

(a) While the cable is mounted in a suitable fixture, a 90-degree bend test shall be performed at minus 40 plus or minus 3 degrees F. The missile control assembly and test fixture shall be conditioned at the test temperature for at least 12 hours prior to initiation of the test.

(b) The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 6 inches shall be formed within a 12-inch length of cable (see Figure 4). The cable shall be rigidly secured to the underside of a fixed support lying in a horizontal plane at one clamp position, and rigidly secured to a pivoting arm which shall be so mounted as to oscillate from a vertical position to a horizontal position. When in its horizontal position, its axis shall form an extension of the plane of the fixed support, and the cable shall be attached to the underside of the arm. The attachment of the cable to the oscillating arm shall be made when the arm is in the vertical position. The pivot axis of the oscillating arm shall intersect the longitudinal axis of the fixed support and shall be equidistant from the two clamping points. The fixture will be so designed that the section of cable between the clamp positions shall not touch any portion of the mechanism or cold box during the test.

(c) One cycle of test shall consist of an oscillation of the pivot arm from the vertical position to the horizontal and back to the vertical. The test shall be conducted at a rate of one cycle per minute, with each half cycle to be accomplished within 6 seconds. The delay between cycles may occur at the end of either half cycle or may be divided to occur at the end of each half cycle.

(d) Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 500 cycles of bending.

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4.5.5 Electrical tests

4.5.5.1 Dielectric strength. The cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6111) to determine conformance with 3.8.1.

4.5.5.2 Insulation resistance. The cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6031) to determine conformance with 3.8.2.

4.5.5.3 Continuity. Each circuit shall be probed with a 6-12 vdc buzzer, or equivalent, for continuity. There shall be no evidence of discontinuity, to meet the requirements of 3.8.3.

4.5.5.4 Contact, conductor resistance. The missile control cable assembly shall be mated to the appropriate torpedo tube cable connector, NAVSEA LD 162874 or LD 162875, and the resistance measurement made from each female socket of the connector to the corresponding male pin in the in-line connector (see Figure 5). The ohmmeter or Kelvin bridge shall be capable of measuring a resistance of less than 0.1 ohm with an accuracy of at least plus or minus 0.02 ohm. The measured resistance of each circuit shall not exceed the values specified in 3.8.4.

4.5.6 Environmental tests

4.5.6.1 Vibration. The vibration test shall be performed with the cable in the mated condition. The mated cable assembly and torpedo tube cable connector shall be mounted to a suitable vibration fixture and this fixture in turn shall be mounted to a sinusoidal vibration system as illustrated in Figure 6. The vibration fixture shall be sufficiently rigid to prevent resonance frequencies greater than 1.5 to 1. All circuits of the mating connectors shall be wired in series. The input control accelerometer shall be placed on the vibration fixture adjacent to the cable tube door mounting point. Two monitoring accelerometers shall be mounted to the cable assembly, one on the in-line connector end and one on the tube door end.

4.5.6.1.1 Vibration schedule and levels. Vibration shall be conducted in three mutually perpendicular directions (see Figure 6) in accordance with the following schedule. All tests in one direction shall be completed before proceeding to tests in another direction.

NOTE: All references to amplitude of vibration refer to maximum displacement of sinusoidal motion from the position of rest, or one half the total excursion.

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(a) Resonance search. The sample shall be vibrated at frequencies from 5 Hz (or lowest obtainable but not higher than 10) to 33 Hz at a table vibratory amplitude of 0.010 plus or minus 0.002 inch and from 34 Hz to 50 Hz at a table vibratory amplitude of 0.003 plus 0 minus 0.001 inch. The change in frequency shall be made in 1-Hz steps and maintained at each step for 15 plus or minus 5 seconds. The frequencies at which the greatest resonance or discontinuities in excess of 100 microseconds occur shall be noted.

(b) Vibration levels. The sample shall be vibrated in accordance with Table IV. The dwell time at each integral frequency shall be 5 minutes.

TABLE IV

Frequency Range (Hz)	Table Amplitude (Inches) (See Note Above)
4 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002
34 to 40	0.005 ± 0.001
41 to 50	0.003 + 0.000 - 0.001

(c) Resonance dwell. The sample shall be vibrated for 2 hours at the frequency at which the greatest resonance or discontinuities were noted. If no discontinuity or resonance occurs at this frequency, this test shall be performed at 50 Hz.

4.5.6.1.2 The above schedule shall be completed in each of the three mutually perpendicular axes. At the conclusion of the vibration tests, the specimens shall be subjected to the tests of 4.5.5.2 and 4.5.5.3.

4.5.6.2 Shock. The missile control cable shall be mated to the appropriate torpedo tube cable connector and securely mounted to a shock fixture simulating service conditions as shown in Figure 6. All circuits shall be wired in series for the purpose of monitoring any discontinuities. The test specimens shall be subjected to the shock test defined in MIL-S-901 for light-weight equipment. Three shocks shall be applied parallel to each of the three principal axes as shown in Figure 6. A total of nine shocks shall be applied. During shock, the series circuit shall be monitored for discontinuity, using a relay chatter indicator or equivalent.

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Discontinuities in excess of 100 microseconds shall be considered evidence of failure.

4.5.6.2.1 At the conclusion of the shock test, the specimens shall be subjected to the tests of 4.5.5.2 and 4.5.5.3.

4.5.6.3 Salt water immersion. The cable assembly, mated with the appropriate tube door connector, shall be subjected to a 30-day immersion test. The cable assembly shall be attached to a fixture so that all of the plug end (tube door end) will be immersed in sea water. The cable assembly shall be immersed for 16 hours and removed for a period of 8 hours. This sequence shall be repeated 30 times. No fresh water rinse will be permitted during the 30 cycles. At the conclusion of this test the connectors shall be subjected to the tests of 4.5.5.2 and 4.5.5.3.

4.5.6.4 Low temperature (operating). The cable assembly shall be mated to the appropriate torpedo tube cable connector and mounted in the test chamber. The chamber temperature shall be lowered to plus 28 degrees F plus or minus 5 degrees F and maintained for a period of 48 hours. At the end of this time and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.2, 4.5.5.2, and 4.5.5.3.

4.5.6.5 High temperature (operating). The cable assembly shall be mated to the appropriate torpedo tube cable connector and mounted in the test chamber. The chamber temperature shall be raised to 130 degrees F plus or minus 5 degrees F and maintained for a period of 48 hours. At the end of this time and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.2, 4.5.5.2, and 4.5.5.3.

4.5.6.6 Temperature and humidity (operating). Both SUBROC umbilical and missile control cables shall be mated to each of two cable connectors. The units shall then be placed in a test chamber and subjected to a temperature of plus 122 degrees F for 2 hours. During this 2-hour period the requirements of 4.5.5.2 and 4.5.5.3 shall be met.

(a) After 2 hours, the receptacles shall be subjected to five 24-hour cycles of 16 hours at plus 122 degrees F and 8 hours at plus 86 degrees F. The relative humidity shall be maintained constant at 95 percent during the five cycles. While at plus 122 degrees F the requirements of 4.5.5.2 and 4.5.5.3 shall be met.

(b) Upon completion of five cycles, the temperature shall be maintained at plus 122 degrees F for 8 hours. During the 8-hour period, the relative humidity shall be 50 percent for the first 2 hours, 95 percent during the next 4 hours, and 40 percent during the final 2 hours. At 30-minute intervals the requirements of 4.5.5.3 shall be met. Upon completion, the requirements of 4.5.2 and 4.5.5.2 shall be met.

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NOTE: The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is approximately 16 hours. Each transition shall not exceed 1½ hours. The relative humidity need not be controlled during the transition periods.

4.5.6.7 Temperature and humidity (storage mode). For this test, the cable assembly, in the packaged condition, shall be subjected to a 28-day temperature/humidity test. The total test is made up of two 14-day temperature and humidity cycles. The term "chamber temperature," as used in this test, refers to the temperature of the air immediately surrounding the equipment. Chamber temperature may be changed (1) by varying the temperature of a single chamber, or (2) by moving the test specimen from one constant temperature chamber to another. The fluctuations from all specified temperatures shall not exceed 10 degrees F. The following test sequence is based on a start at 0800 Monday. Regardless of the actual day and time the test is initiated, there shall be no deviation from the test sequence as prescribed.

(a) At 0800 Monday place the test specimen in a chamber which has been maintained at minus 65 degrees F for at least 2 hours immediately prior to 0800 (humidity not controlled).

(b) At 1600 Monday the chamber temperature shall be changed to plus 160 degrees F and 95 plus 3 minus 0 percent humidity. The chamber temperature must reach 160 degrees F and 95 percent humidity not later than 1800.

(c) The test specimens shall be held under these conditions until 0800 Tuesday, at which time the temperature decrease shall begin. The rate of decrease of chamber temperature shall be equal to or greater than 35 degrees F per hour for at least 2½ hours. Thus, at 1030, the temperature of the air surrounding the test specimens must be plus 70 degrees F or lower. The chamber temperature must reach minus 65 degrees F no later than 1400 and this temperature shall be held until 1600.

(d) At 1600 Tuesday the chamber temperature shall be changed to plus 160 degrees F and 95 plus 3 minus 0 percent humidity as rapidly as practicable (not later than 1800) and held until 0800 Wednesday.

(e) On Wednesday, Thursday, and Friday the operations performed on Tuesday shall be repeated.

(f) The chamber temperature shall be raised to plus 160 degrees F and 95 percent humidity on Friday evening and shall be so maintained until 0800 Monday.

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(g) At 0800 Monday the sequence of operations described above for Tuesday of the first week shall be followed and shall be repeated daily until Friday of the second week.

(h) On Friday, the chamber temperature shall be reduced to minus 65 degrees F, and this temperature shall be maintained until 0800 Monday of the third week. The cycle will be complete at 0800 Monday.

4.5.6.7.1 The above sequence of temperature and humidity conditions will constitute one cycle. Two such cycles shall be applied. The second cycle shall be completed at 0800 Monday of the fifth week. At the time, the test specimen shall be allowed to return to room temperature. Without drying, wiping, or baking, the test specimen shall meet the requirements of 4.5.5.2 and 4.5.5.3.

4.5.6.8 Tropical environment. For this test, the cable assembly shall be mated to the appropriate torpedo tube cable connector. The coupled unit shall be mounted in the test chamber wall or suitably protected so as to prevent exposing the in-line connector face and the tube side of the connector.

(a) The test specimen shall be placed in a mold chamber in which a temperature of 86 degrees F plus or minus 5 degrees F and a relative humidity of 90 percent plus or minus 5 percent is maintained. The test specimen, while so housed, shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for a period of 28 days. At the end of this period, the test specimen shall be allowed to return to room temperature, and it shall be examined for evidence of being nutrient to fungi. If the test specimen shows susceptability to fungi, colored photographs shall be taken.

(b) At least five groups of fungi shall be used in test specimen inoculation. Five such groups are listed in MIL-T-18404 and one type of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at 77 degrees F) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall then be introduced directly into a stock culture and shaken vigorously, in order that a well sporulated suspension will result without disturbing the agar.

(c) This process shall be repeated for each type fungus. The separate spore suspension from the various types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an 8-hour period.

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(d) The cable assembly shall show no evidence of being nutrient to fungi and shall meet the requirements of 4.5.5.2 and 4.5.5.3.

4.6 Packaging, packing, and marking. Packaging, packing, and marking shall be examined for compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The connectors shall be protected with a plastic dust cap and be wrapped in a transparent bag conforming to characteristics equal to or better than a 0.003-inch wall thickness of Type-1, Class-E, Style-2 of MIL-B-117. Loose component parts shall be protected from damage and may be enclosed in similar plastic bags. The cable assembly and components shall be packed in 3/4-inch thick cellulosic material conforming to PPP-C-843, Type-II, Class-B and packed in a 12¼ x 12¼ x 6-13/16 inch fiberboard container in accordance with Drawing 757989.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The cable assembly shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The cable assemblies, packaged as described in 5.1.1.1, shall be packed, three each, in a wooden box in accordance with PPP-B-621, Class-2 and may be shipped individually in small lots.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 91.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The cable assemblies packaged in accordance with 5.1.3 shall be packed to afford protection from damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

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5.3 Marking

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The cable assembly covered by this specification is intended for use aboard submarines in transmitting fire control electrical signals from the torpedo tube door to the fire control console.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type cable assembly required (LD 162872 or LD 162873).

(c) Assigned activity for preproduction inspection.

(d) Applicable levels of packaging (see 5.1).

(e) Applicable levels of packing (including palletizing) (see 5.2).

6.2.1 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.4.1	Certified inspection data	-

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project Number
1355-N363

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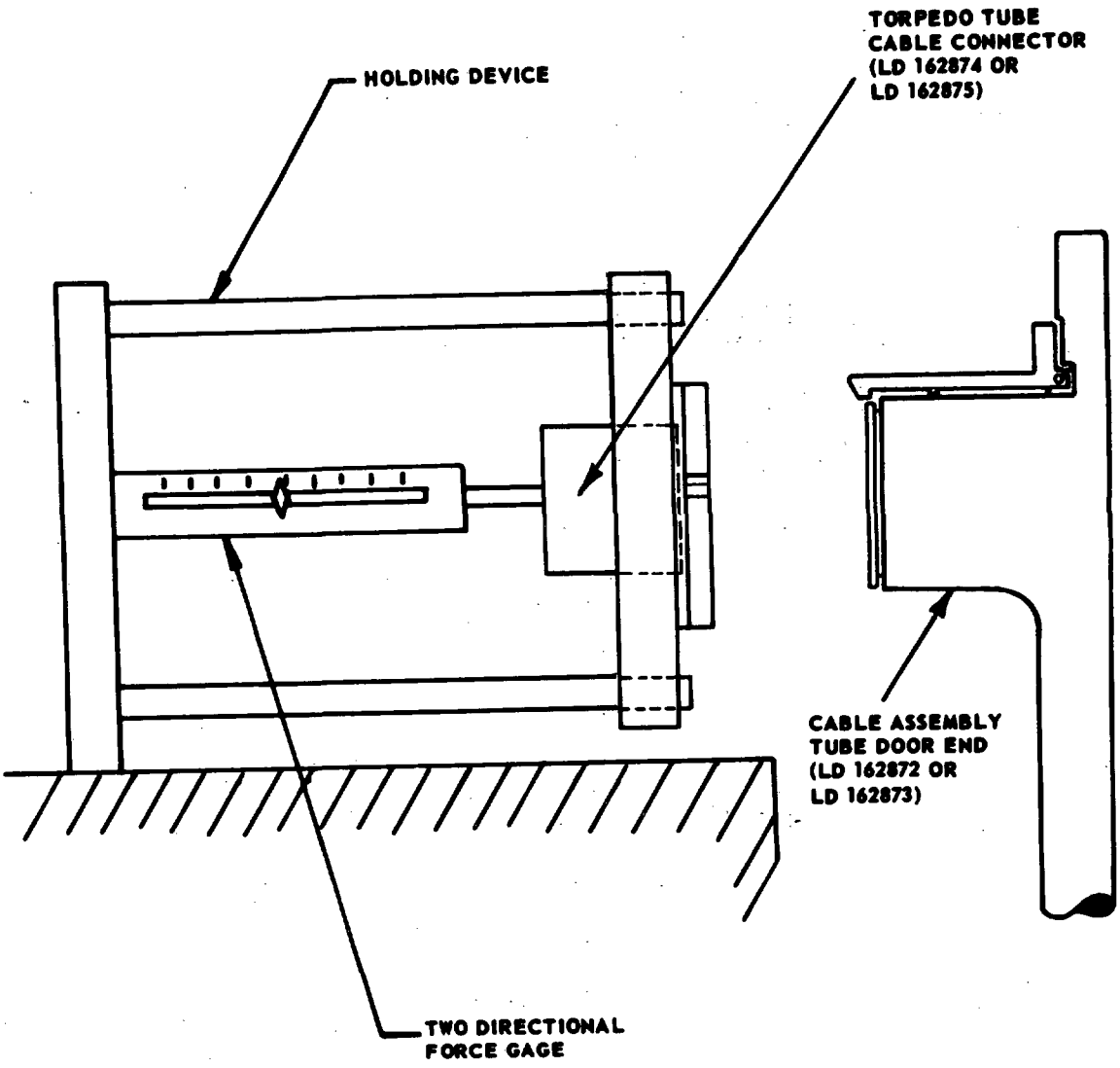


FIGURE 1. TEST SETUP – MATING AND UNMATING FORCE.

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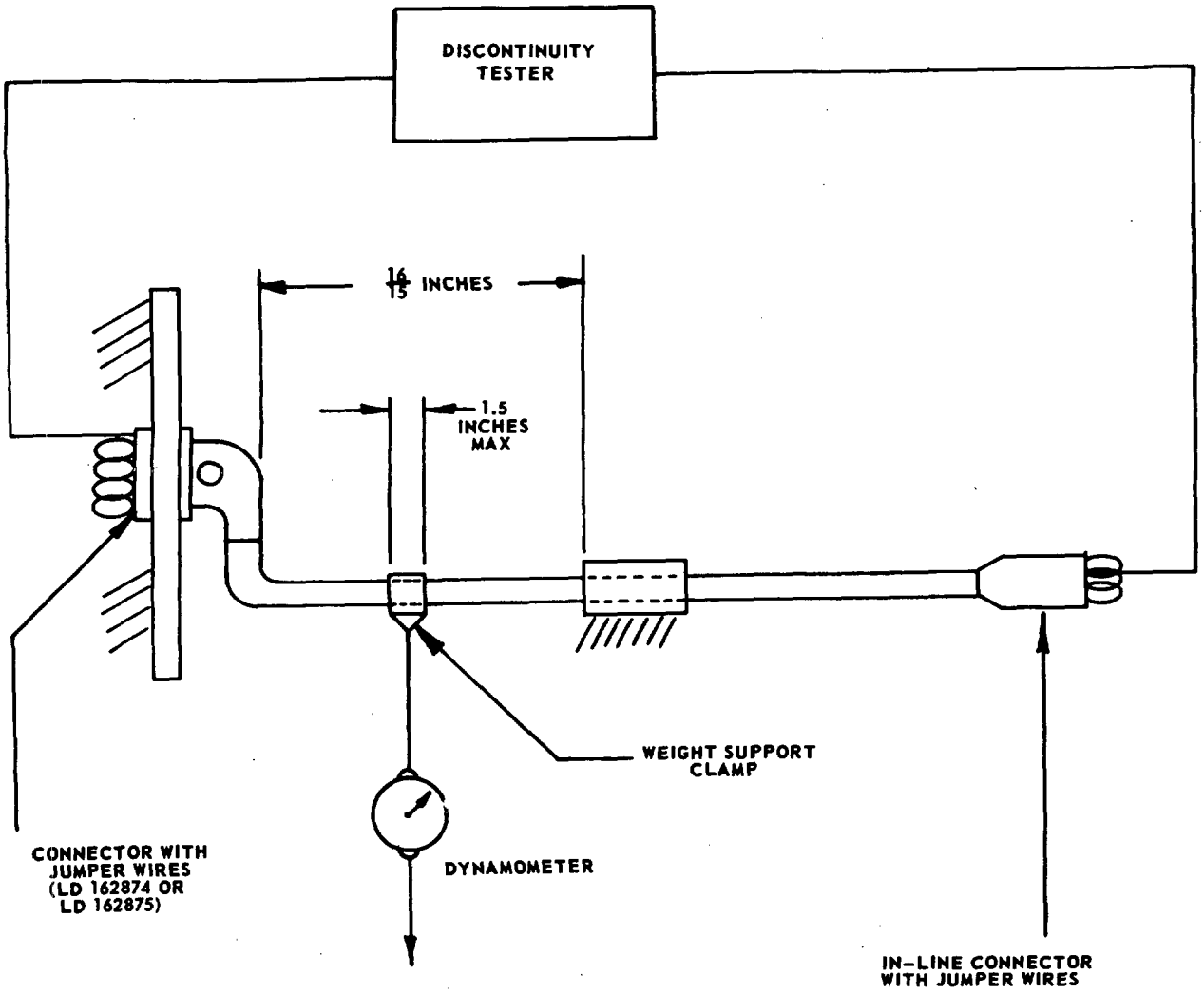


FIGURE 2. TEST SETUP – CABLE STRENGTH.

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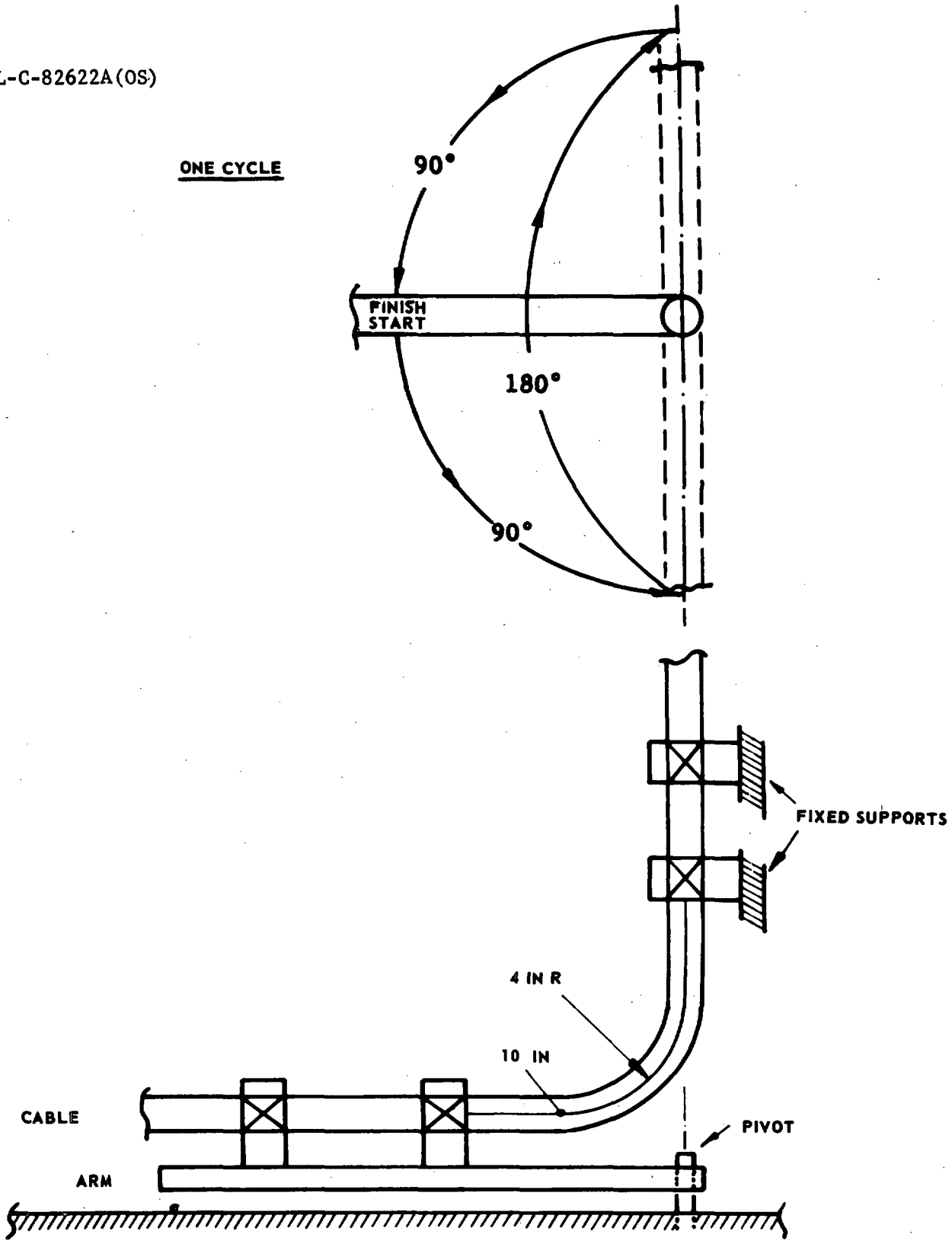


FIGURE 3. TORSION TEST SETUP.

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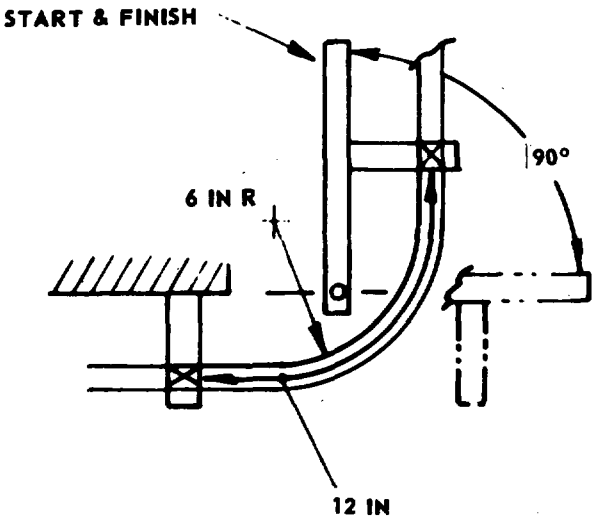


FIGURE 4. TEST SETUP – 90-DEGREE BEND LOW TEMPERATURE.

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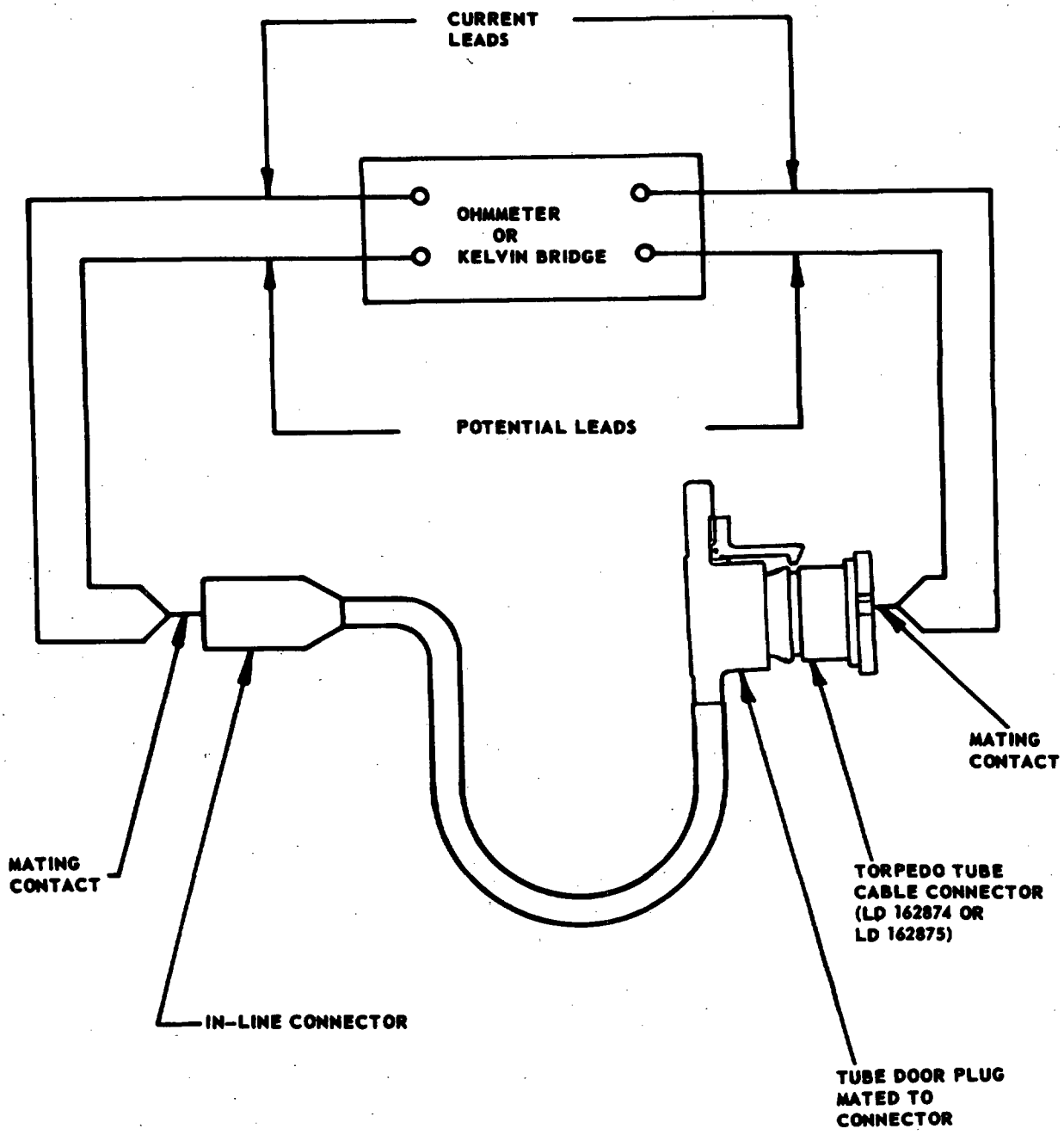


FIGURE 5. TEST SETUP – CIRCUIT RESISTANCE.

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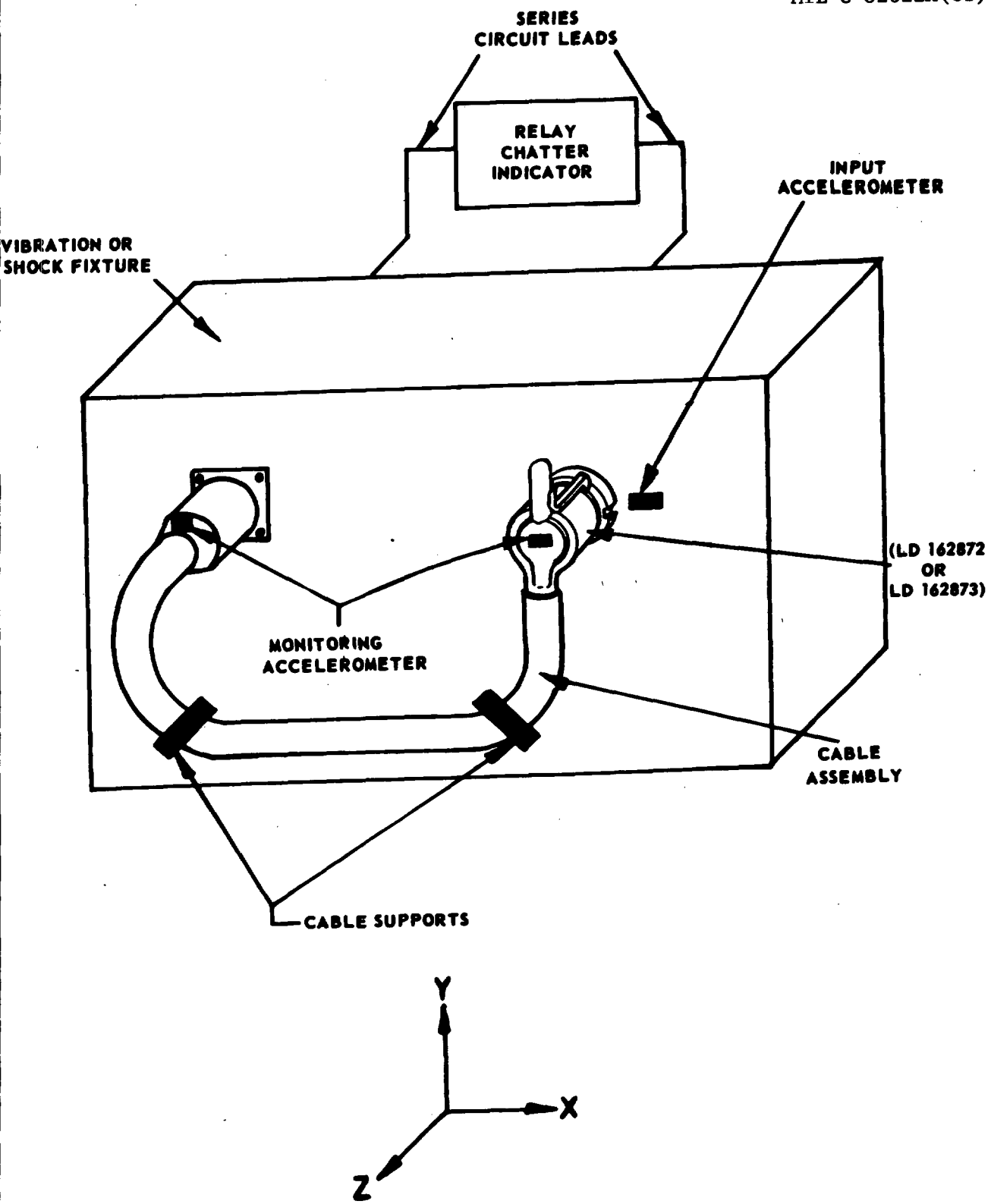


FIGURE 6. TEST SETUP – VIBRATION AND SHOCK.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-82622A(OS) Cable, Missile Control, Torpedo Tube, Mk 8 Mod 0 and Mk 9 Mod 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standardization Division (502)
Indian Head, MD 20640

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